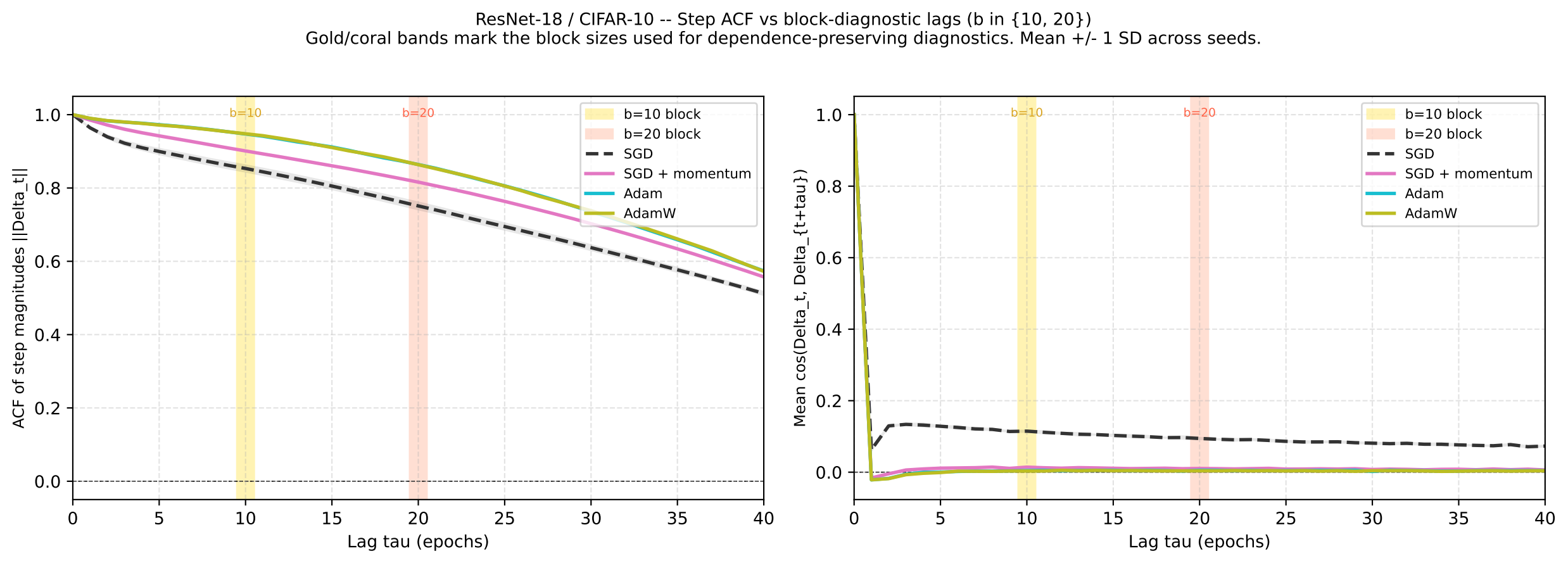
ResNet-18 / CIFAR-10 -- Step ACF vs block-diagnostic lags (b in {10, 20})
Gold/coral bands mark the block sizes used for dependence-preserving diagnostics. Mean +/- 1 SD across seeds.
ACF of step magnitudes ||Delta_t||
Lag tau (epochs)
b=10
b=20
b=10 block
b=20 block
SGD
SGD + momentum
Adam
AdamW
Mean cos(Delta_t, Delta_{t+tau})
Lag tau (epochs)
b=10
b=20
b=10 block
b=20 block
SGD
SGD + momentum
Adam
AdamW